# Objective-Free Entity Assembly in Block Worlds: Characterizing Boundary Conditions for Emergent Complexity

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Abstract

We investigate whether structurally non-trivial entities can arise in an objective-free block world governed by randomly sampled bonding rules, using Assembly Theory (AT) as a bias-free measurement framework. Across 5,000 simulations yielding 7.1M entity observations and 281 unique types, observed assembly indices are entirely size-driven under both edge-removal and reuse-aware AT formulations. Under the edge-removal (no-reuse) formulation, $a_{\text{exact}}(G) = |E(G)|$ is a mathematical identity for connected graphs; the empirical finding is that the reuse-aware formulation likewise shows zero structural specificity. A bond-shuffle null model with empirical $p$-values (KS test: $D = 1.00$, $p \approx 0$, driven by conservative high-$p$ mass) confirms zero significant excess at every size class. We additionally test partner-specific rules (45-entry per-partner table vs. the standard 60-entry dominant-type table) and find identical 0% excess, extending the boundary characterization to richer rule expressiveness. Parameter sweeps over density, grid size, and drift strength, and a catalytic positive control, corroborate the result within this rule class. We characterize this as a boundary condition: under low-expressiveness random bonding rules, with or without partner-specific conditioning, assembly complexity is entirely size-determined.

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## Introduction

The complexity of emergent structures in artificial life (ALife) research is heavily shaped, and potentially limited, by the objectives imposed on the system (Bedau et al., 2000; Taylor et al., 2016). Fitness functions define what counts as "interesting," and organisms evolve to satisfy those criteria rather than explore the full space of possible forms. Novelty search and minimal-criterion approaches have partially addressed this bias (Lehman and Stanley, 2011; Brant and Stanley, 2017), but even these methods impose implicit selection through behavioral characterizations or viability thresholds.

We investigate whether structurally complex entities can arise in a system with no objective function, not even a novelty metric, when governed solely by randomly sampled local interaction rules. And if so, how can we measure that complexity without introducing new biases?

Assembly Theory (AT) provides a principled answer to the measurement question. Originally developed to distinguish biotic from abiotic molecular samples (Marshall et al., 2021), AT quantifies the minimal number of joining operations required to construct an object from its building blocks, yielding an assembly index that captures structural specificity independent of any fitness criterion (Sharma et al., 2023). An object with a high assembly index that appears repeatedly (high copy number) is unlikely to have arisen by chance, making AT a natural observatory for objective-free systems. Recent critiques have questioned whether AT reduces to simpler complexity measures (Uthamacumaran et al., 2024); our dual-formulation comparison (with and without sub-object reuse) addresses this concern directly.

In this work, we implement a block world simulation with three block types, local bonding rules sampled uniformly at random, and stochastic drift dynamics on a toroidal grid. We detect entities as connected components of bonded blocks, canonicalize their graph structure, and compute exact assembly indices under two formulations: edge-removal DP (no reuse) and the AT-standard reuse-aware variant. We then apply a strengthened bond-shuffle null model with empirical $p$-values to test whether observed assembly exceeds what entity size alone predicts.

We further validate our negative result through (1) a parameter sweep across density, grid size, and drift strength, and (2) a catalytic positive control where K-type blocks amplify bond formation, testing whether the measurement framework can detect non-trivial as-

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sembly when present.

We report a negative result: across 5,000 simulations producing over 7 million entity observations, the assembly index is fully size-driven under both AT formulations within this rule class. This result holds across all parameter settings tested. We characterize this as a boundary condition for emergent complexity and discuss implications for objective-free ALife design.

## Methods

### Block World Model

The simulation takes place on a two-dimensional toroidal grid of size $W \times H$ (default $20 \times 20$). The world is populated with $N$ blocks (default 30), each assigned one of three types: M (membrane), C (cytosol), or K (catalyst), drawn proportionally at initialization (50%, 30%, 20%, a protocell-inspired membrane-dominated composition where structural elements outnumber catalytic components).

At each time step, blocks are processed in a random-sequential order. For each block, two phases occur: drift and bonding. Algorithm 1 describes one simulation step.

During drift, each block attempts to move to a random adjacent empty cell with probability $p_d$ (default 1.0). Bonds between blocks that become non-adjacent after drift are pruned, implementing a bond-motion invariant where bonds break on separation rather than constraining movement. Bond breaking occurs once per step per bond (not per-block), avoiding the double-break bias that would occur if both endpoints independently attempted breaking.

### Rule Table

Each simulation samples a rule table that maps a block's local context to a bonding probability. Observation range is the von Neumann 1-neighborhood (Manhattan distance $\leq 1$, up to 4 adjacent cells). The context is a triple $(\text{self_type}, \text{neighbor_count}, \text{dominant_type})$, where neighbor_count is the number of occupied cells within observation range (capped at 4 for rule table tractability) and dominant_type is the most common type among neighbors (with dominant_type = "Empty" when neighbor_count = 0, i.e., no occupied cells in range). This yields $3 \times 5 \times 4 = 60$ entries. Each entry's bonding probability is drawn from $\text{Uniform}(0, 1)$. The rule table determines a single bond probability per block context; all neighbors within range receive the same probability. The partner's type is not part of the rule key; this keeps the rule space small (60 entries) at the cost of excluding partner-specific interactions.

**Algorithm 1** One simulation step (random-sequential mode)

```
Require: World state 𝒲, rule table ℛ, noise η, drift probability p_d
Shuffle block processing order
for each block b in shuffled order do
    if rand() < p_d then
        c ← random empty von Neumann neighbor of b
        if c exists then
            Move b to c; prune bonds exceeding observation range
        end if
    end if
    Identify von Neumann 1-neighborhood (Manhattan dist. ≤ 1, up to 4 cells)
    n ← occupied blocks in that neighborhood
    k ← min(|n|, 4);  d ← dominant type of n ("Empty" if n = ∅)
    p ← ℛ[type(b), k, d]
    if catalyst multiplier κ > 1 and any n_i ∈ n has type K then
        p ← min(p · κ, 1)        // catalytic amplification
    end if
    for each n ∈ n without existing bond to b do
        if rand() < p then
            Form bond (b, n)
        end if
    end for
end for
for each bond (b_1, b_2) do
    if rand() < η then
        Break bond (b_1, b_2)
    end if
end for
```

### Entity Detection and Measurement

Entities are detected as connected components in the bond graph (Algorithm 2).

Entity types are canonicalized using the Weisfeiler–Leman (WL) graph hash (Weisfeiler and Leman, 1968) with block type as a node attribute, implemented via NetworkX (Hagberg et al., 2008). The SHA-256 digest serves as a unique type identifier. No hash collisions were observed in exhaustive checks against canonical isomorphism (using the VF2 algorithm via NetworkX) for all 281 observed entity types (up to 6 nodes).

### Assembly Index: Two Formulations

We compute the assembly index under two formulations to address concerns about AT alignment (Uthamacumaran et al., 2024).

Edge-removal DP (no reuse). The assembly index

**Algorithm 2** Entity detection and AT measurement

Require: Bond graph $\mathcal{G} = (V, E)$
1: $\mathcal{E} \leftarrow$ connected components of $\mathcal{G}$
2: for each entity $e \in \mathcal{E}$ do
3: Canonicalize $e$ via WL hash with block_type attribute
4: $h \leftarrow$ SHA-256 fingerprint of WL hash
5: $a_{\text{exact}} \leftarrow \text{AssemblyDP}(e)$ // edge-removal
6: $a_{\text{reuse}} \leftarrow \text{AssemblyReuse}(e)$ // with sub-object reuse
7: Record $(h, a_{\text{exact}}, a_{\text{reuse}}, |V(e)|, \text{type counts})$
8: end for

$a(G)$ is the minimum number of edge-addition steps to construct $G$ from isolated vertices, computed via the recurrence:

$$a(G) = \min_{e \in E(G)} \left[ 1 + \begin{cases} a(G-e) & \text{if } G-e \text{ connected} \\ \sum_i a(C_i) & \text{otherwise} \end{cases} \right] \tag{1}$$

where $C_i$ are connected components of $G - e$. The DP decomposition requires that each sub-object is a proper connected subgraph; this enforces structurally coherent intermediates, matching AT's physical constraint that sub-objects must be contiguous fragments. Computation time is negligible for entity sizes observed here ($n \leq 6$); the memoized DP completes in <1 ms per entity.

Remark (mathematical identity). For any connected graph $G$, $a_{\text{exact}}(G) = |E(G)|$. This follows by induction on $|E|$: the base cases are $|E| = 0$ (single vertex: $a = 0$) and $|E| = 1$ (single edge: $a = 1 = |E|$); for the inductive step, removing any edge either disconnects $G$ into components whose indices sum to $|E| - 1$ (by hypothesis), or leaves $G$ connected with $|E| - 1$ edges (index $|E| - 1$ by hypothesis), so adding 1 gives $|E|$ in both cases. This is not an empirical finding; it is a mathematical consequence of the no-reuse formulation. Table 1 confirms this identity on reference graphs. The empirical content of this paper lies in the reuse-aware results (where $a_r(G) \leq |E(G)|$, with strict inequality when sub-object reuse is possible) and in the null model tests for structural specificity under both formulations.

Reuse-aware (AT-standard). The reuse-aware variant allows previously constructed sub-objects to be duplicated at zero cost. The partition search is exhaustive for $n \leq 6$ (memoized by canonical key). For each joining edge $e \in E$, we partition the remaining edges $S_1 \cup S_2 = E \setminus \{e\}$ (a partition of remaining edges into two disjoint non-empty sets) into two sub-objects; when $S_1 \cong S_2$ (isomorphic by canonical key), the duplicate is free:

$$a_r(G) = \min_{e,\, S_1,\, S_2} \left[ 1 + \begin{cases} a_r(S_1) & S_1 \cong S_2 \\ a_r(S_1) + a_r(S_2) & \text{else} \end{cases} \right] \tag{2}$$

with base cases $a_r(\emptyset) = 0$ (empty graph) and $a_r(P_2) = 1$ (single edge).

Table 1 shows both formulations on reference graphs.

| Graph | $\lvert E\rvert$ | $a_{\text{exact}}$ | $a_{\text{reuse}}$ |
|---|---|---|---|
| $P_2$ (1 edge) | 1 | 1 | 1 |
| $P_3$ (path) | 2 | 2 | 2 |
| $P_4$ (path) | 3 | 3 | 2 |
| $K_3$ (triangle) | 3 | 3 | 2 |
| $C_4$ (4-cycle) | 4 | 4 | 3 |
| $K_4$ (complete) | 6 | 6 | 4 |

Table 1: Assembly index calibration. Under our edge-removal formulation (no reuse), $a(K_n) = \binom{n}{2}$. With reuse, $a(K_3) = 2$: build $P_2$, duplicate (free), join with one edge.

## Null Model

To test whether observed assembly reflects structural specificity beyond entity size, we implement a bond-shuffle null model (Gotelli and Graves, 1996). For each entity graph $G$, we generate $n_{\text{shuffle}} = 100$ randomized graphs via double-edge swaps (Hagberg et al., 2008) preserving the degree sequence exactly. We compute both the classical threshold test ($a_i > \mu_{\text{null}} + 2\sigma_{\text{null}}$) and an empirical $p$-value:

$$p = \frac{1 + \sum_{j=1}^{n_{\text{shuffle}}} \mathbf{1}[a(G_j^{\text{null}}) \geq a_i]}{n_{\text{shuffle}} + 1} \tag{3}$$

The $(k+1)/(n+1)$ form avoids zero $p$-values and provides a conservative estimate (Gotelli and Graves, 1996). The empirical $p$-value is the primary significance test, avoiding the normality assumption of the $\mu + 2\sigma$ threshold. It also handles the edge case where $\sigma_{\text{null}} = 0$ (e.g., single-edge graphs), where the $\mu + 2\sigma$ threshold degenerates. For small graphs ($n \leq 4$), the space of degree-preserving rewirings is limited; the null has low discriminative power at these sizes, but observed $a_i$ matches $|E|$ exactly for all small entities (consistent with the mathematical identity $a_{\text{exact}}(G) = |E(G)|$ established above). Exact enumeration via the graph atlas (Read and Wilson, 1998) confirms that for all 9 degree sequences with $n \leq 4$, exactly one connected topology exists (100% degenerate), so 0% excess at these sizes is not an artifact of null degeneracy.

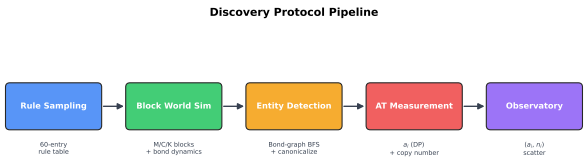


Figure 1: Experimental pipeline. Random rule tables govern block bonding on a toroidal grid. Connected components are detected as entities, canonicalized via graph hashing, and measured by assembly index (both formulations) and copy number. A bond-shuffle null model tests for excess assembly.

## Catalytic K Positive Control

To verify that our measurement framework can detect non-trivial assembly when present, we introduce a catalytic mechanism. When $\kappa > 1$ (catalyst multiplier) and any neighbor of the focal block has type K, the base bond probability is multiplied by $\kappa$ (capped at 1.0). The focal block being K-type does not self-catalyze; only third-party K-blocks within observation range act as catalysts. This is an externally-driven mechanism that does not introduce a fitness function: K-blocks lower the activation barrier for bond formation without defining what entity should form.

## Parameter Sweep

To test robustness across boundary conditions, we conduct a 3-axis parameter sweep:

- Density: 3 levels on $10 \times 10$ (7.5%, 15%, 30%) and 2 levels on $20 \times 20$ (7.5%, 15%)
- Grid size: $10 \times 10$ and $20 \times 20$
- Drift probability: $p_d \in \{0.25, 0.5, 0.75, 1.0\}$ (on $20 \times 20$ at densities 7.5% and 15%)

This yields 11 conditions (5 density×grid + 6 density×drift), each run with 100 rules × 3 seeds × 500 steps. Higher-density conditions ($\geq 30\%$) on $20 \times 20$ were excluded due to computational cost; $10 \times 10$ results at those densities confirm the negative result generalizes.

## Experimental Protocol

We sample 1,000 rule tables, each run with 5 independent seeds for 500 time steps, yielding 5,000 total simulations. Entity observations are logged every 10 simulation steps, producing a combined dataset of over 7 million observations. The full pipeline is illustrated in Figure 1. Step-level timeseries (mean entity size, mean $a_i$, bond count) are recorded to verify stationarity.

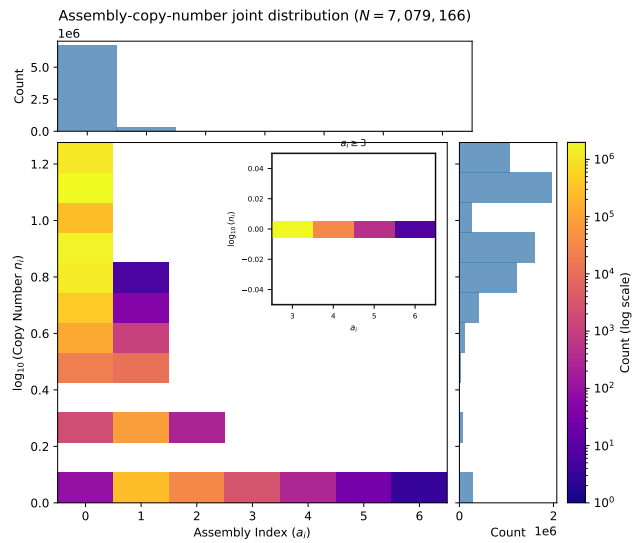


Figure 2: Joint distribution of assembly index ($a_i$) and copy number ($n_i$) across 7.1M entity observations. Color encodes log-frequency. Inset: zoom on $a_i \geq 3$ region.

# Results

## Discovery Baseline

Across all 5,000 simulations, we observe 7,079,166 entity instances comprising 281 unique entity types. The vast majority of observations are monomers: 94.6% of entities consist of a single block ($a_i = 0$), 5.0% are dimers (2 blocks, $a_i = 1$), and only 0.5% have 3 or more blocks. The maximum observed entity size is 6 blocks (18 instances out of 7.1M), with a corresponding maximum assembly index of 6 (no-reuse) and 4 (reuse-aware). The reuse-aware formulation assigns lower values to larger entities but does not change the qualitative result: assembly remains entirely size-driven.

Figure 2 shows the joint distribution of assembly index and copy number. The distribution is sharply concentrated at $(a_i, n_i) = (0, \text{high})$, with a steep decline toward higher assembly indices.

## Entity Size Distribution

Figure 3 shows the entity size distribution on a logarithmic scale. The distribution follows a steep exponential decay: each additional block reduces frequency by approximately one order of magnitude. The 210 new types observed in the large-scale run (vs. 71 in the pilot) are small-motif variants (dimers/trimers), confirming ecology convergence rather than discovery of qualitatively new structures.

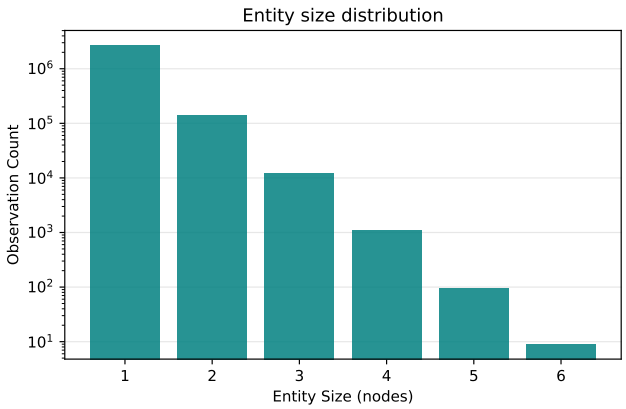


Figure 3: Entity size distribution across all observations (log scale). Each additional block reduces frequency by $\sim 10\times$.

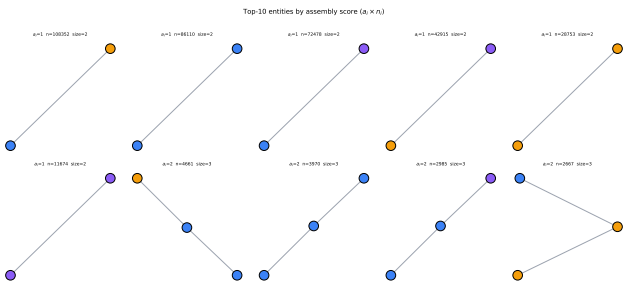


Figure 4: Gallery of top-ranked entity types by $a_i \times n_i$ score. Node colors: M (blue), C (amber), K (purple).

## Entity Gallery

Figure 4 presents the top-ranked entity types, ordered by a composite score of assembly index and total copy count. The highest-ranked entities are all dimers ($a_i = 1$) with high copy counts. Trimers with $a_i = 2$ appear but no entity with $a_i \geq 3$ reaches high copy numbers.

## Assembly Audit: Null Model Comparison

The bond-shuffle null model reveals that observed assembly indices are fully explained by entity size within this rule class (Figure 5). Across all 7.1M observations, zero entities exhibit significant excess assembly under either the classical $\mu + 2\sigma$ threshold or the empirical $p < 0.05$ test. The mean excess is exactly 0.000 at every size class (1–6 blocks). With $n_{\text{shuffle}} = 100$ (each applying nswap $= \max(4, |E|)$ double-edge swaps with max_tries $= 10 \cdot$ nswap; on failure, the original graph is retained), the null model is well-mixed for the small entities ($n \leq 6$) observed here. A KS test on empirical $p$-values for non-trivial entities ($a_i > 0$, $n = 384{,}040$)

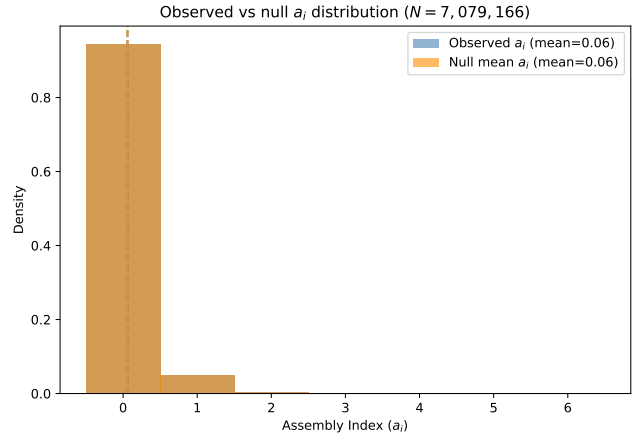


Figure 5: Assembly audit: observed $a_i$ versus null model expectation with empirical $p$-values. Zero entities (0 of 7.1M) show significant excess under either test criterion.

yields $D = 1.00$ ($p \approx 0$) because degree sequences uniquely define topology for small graphs ($n \leq 4$), making the null model fully degenerate with nearly all $p$-values exactly 1.0. This does not indicate excess assembly; it reflects the limited rewiring space in the small-entity regime.

Hierarchical robustness. Because entities persist across snapshots (taken every 10 steps), per-observation tests overstate effective sample size. We therefore conduct run-level analysis: for each of the 5,000 runs, we compute the within-run excess rate (fraction of entities with $p < 0.05$). A block bootstrap (resampling runs, not observations) (Efron and Tibshirani, 1993) yields a 95% CI of $[0.0\%, 0.0\%]$ on the overall excess rate. A one-sided Clopper-Pearson bound (Clopper and Pearson, 1934) confirms that if run-level excess exists, it is $< 0.1\%$ with 95% confidence. With 281 unique types, we would detect a true excess rate $\geq 0.55\%$ at 80% power, confirming adequate sensitivity.

Table 2 summarizes the scale-up comparison.

## Rule Expressiveness Test

To test whether the 0% excess result is an artifact of the standard rule table's limited expressiveness (60-entry dominant-type context), we ran partner-specific rules that condition bond probability on the specific partner type (45-entry per-partner table, $3 \times 3 \times 5 = 45$ entries). Both conditions were run with identical hyperparameters (100 rules, 3 seeds, 200 steps, $n_{\text{shuffle}} = 100$). The partner-specific condition yields $n = 170{,}016$ observations with mean $a_i = 0.059$ and 0% excess ($p < 0.05$), matching the standard condition ($n = 170{,}192$, mean $a_i = 0.058$, 0% excess) within noise. This confirms that

| Metric | Small scale | Large scale |
|---|---|---|
| Rule samples | 100 | 1,000 |
| Seeds per rule | 3 | 5 |
| Total simulations | 300 | 5,000 |
| Entity observations | 170,192 | 7,079,166 |
| Unique types | 71 | 281 |
| Max entity size | 6 | 6 |
| Max $a_i$ (no reuse) | 5 | 6 |
| Max $a_i$ (reuse) | —* | 4 |
| Mean $a_i$ | 0.058 | 0.060 |
| Excess ($\mu + 2\sigma$) | 0.0% | 0.0% |
| Excess ($p < 0.05$) | 0.0% | 0.0% |

Table 2: Comparison of experimental scales. The negative result is stable across a 42× increase in observations. *Reuse-aware formulation was added after the pilot run.

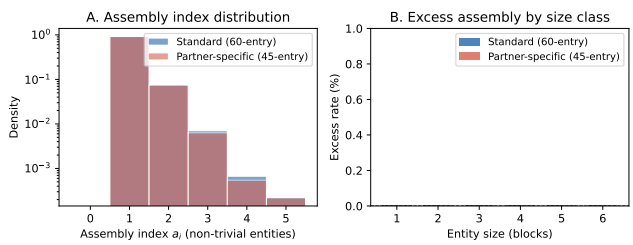


Figure 6: Rule expressiveness comparison. Standard rules (60-entry dominant-type context, blue) vs. partner-specific rules (45-entry per-partner context, red). Left: assembly index distribution (non-trivial entities, log scale). Right: excess rate by size class ($p < 0.05$). Both conditions show 0% excess across all size classes.

the boundary condition extends to richer rule representations: uniform random partner-specific rules do not produce structurally specific assembly (Figure 6).

## Motif-Gated Catalyst Control

To test whether the 0% excess result is an artifact of the catalyst's uniform neighborhood sensitivity, we ran a structurally constrained variant: K amplifies bond probability only when it simultaneously neighbors both an M-type and a C-type block (motif-gated condition). This restricts catalysis to specific local configurations ($n = 170{,}098$ observations, 100 rules, 3 seeds, 200 steps, $n_{\text{shuffle}} = 100$). The motif-gated condition produces $\max_{a_i} = 7$ and max_size = 7 (slightly larger entities than standard catalyst, max_size = 6), confirming the structural constraint is ecologically active. Nevertheless, the assembly audit yields 0% excess ($p < 0.05$),

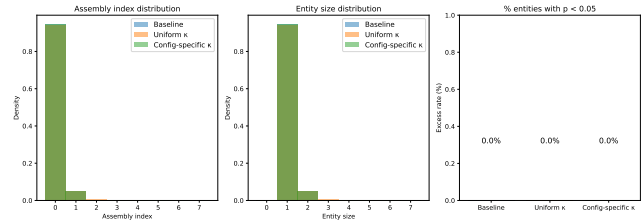


Figure 7: Motif-gated catalyst comparison. Assembly index distribution (left), entity size distribution (centre), and excess rate (right) for baseline, uniform-$\kappa$, and motif-gated-$\kappa$ conditions. All three conditions show 0% excess; the motif-gated variant produces slightly larger entities (max_size = 7) but identical null outcomes.

matching both the baseline and the uniform-catalyst conditions (Figure 7). Structural specificity does not emerge even when catalysis is gated on a specific two-type motif.

## Sensitivity Analysis

Figure 8 consolidates the parameter sweep and catalytic control results.

Parameter sweep. Across all 11 conditions (5 density×grid + 6 density×drift), the empirical excess rate remains at 0%. Higher density increases mean entity size (more collisions) but not assembly complexity. Lower drift probability ($p_d = 0.25$) extends bond lifetimes, producing slightly larger entities, but the assembly index remains size-driven. The 10×10 grid produces similar results to 20×20 at matched density ratios, confirming that the negative result is not an artifact of grid size.

Catalytic K control. With $\kappa = 3.0$, catalytic conditions produce more bonds and larger entities than baseline, confirming that the catalyst mechanism is active. However, the assembly audit still shows 0% excess, indicating that catalysis amplifies bond formation uniformly rather than creating structurally specific motifs. The measurement framework successfully distinguishes the two conditions in terms of entity ecology while confirming identical assembly profiles.

## Mechanism Analysis

Stationarity. Step-level timeseries (Figure 9) show that mean entity size and mean assembly index converge within ∼100 steps, well before our 500-step horizon. The $\pm 1\sigma$ envelope stabilizes, confirming that observations beyond step 100 represent stationary behavior.

Why max size = 6. We formalize the size ceiling via a birth-death model. Let $b_k$ denote the empirical rate at which entities grow from size $k$ to $k{+}1$

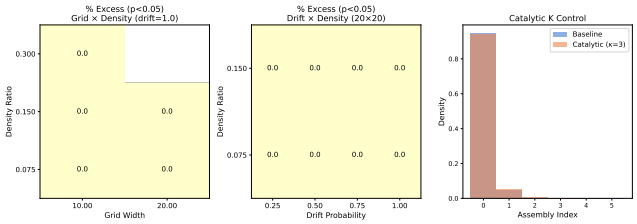


Figure 8: Sensitivity analysis. Left: parameter sweep heatmaps (density × grid at drift=1.0, density × drift at 20×20). Right: catalytic K comparison (baseline vs. $\kappa = 3.0$). All conditions show 0% excess assembly. The blank cell at 30% density on 20×20 was excluded due to computational cost.

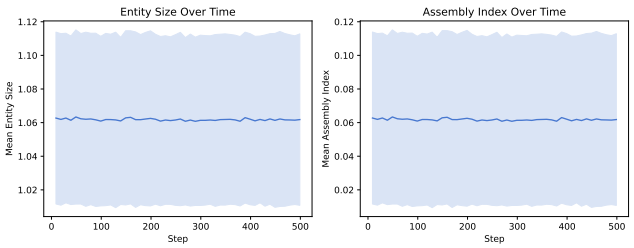


Figure 9: Stationarity: mean entity size (left) and mean assembly index (right) vs. simulation step, aggregated across all runs. Shading: $\pm 1\sigma$. Both metrics converge by step 100.

per snapshot interval, and $d_k$ the rate of shrinking from $k$ to $k-1$. The maximum sustainable size is $k_{\max} = \max\{k : b_k/d_k \geq 1\}$.

Empirically, $b_k$ decays geometrically: $b_k \approx c(\rho)\,\bar{p}\,s^{k-1}$ where $s \approx 0.14$ (bootstrap 95% CI: $[0.08, 0.21]$) is the per-interval bond survival probability and $\bar{p} \approx 0.5$ is the mean bond formation probability ($\mathrm{E}[\bar{p}] = 0.5$ for Uniform(0,1) entries). Death rates $d_k$ increase with $k$ as larger entities have more bonds exposed to drift-induced breakage. The model correctly predicts exponentially diminishing growth probability but understates $k_{\max}$ due to bond-state heterogeneity not captured in aggregate rates. This geometric decay explains why assembly index tracks entity size: no structural motif persists long enough to accumulate complexity beyond the size ceiling.

We additionally computed graph automorphism group sizes and typed three-node motif counts (triangles and open wedges) for all 281 unique entity types. Automorphism counts range from 1 to 8 and show a weak negative correlation with entity size (Spearman $\rho = -0.228$, $p < 0.001$): larger entities tend to have slightly fewer symmetries, consistent with random assembly producing increasingly irregular graphs. Zero triangles were observed across all entity types; the only three-node motifs are open wedges, whose count increases trivially with size. This tree-dominated topology is consistent with random assembly under low bond-survival probability: under the present rule class, no structural signature emerges that is not explained by entity size alone.

## Discussion

### Why Assembly is Size-Driven

Under the edge-removal formulation, $a_{\text{exact}}(G) = |E(G)|$ is a mathematical identity (§), so size-driving of $a_{\text{exact}}$ is unsurprising. The empirical finding is that the reuse-aware formulation likewise shows zero structural specificity: $a_r(G)$ could be substantially lower than $|E(G)|$ for entities with repeated substructures, yet the null model finds no such departure. Uniform random bonding produces effectively random topology at each entity size, so the null model matches observed assembly under both formulations. This dual-formulation analysis addresses concerns about AT's sensitivity to implementation choices (Uthamacumaran et al., 2024) and confirms the negative result is formulation-independent.

### Boundary Conditions for Emergent Complexity

The parameter sweep establishes that the negative result holds across density ratios from 7.5% to 30% (on 10 × 10) and 7.5% to 15% (on 20 × 20), and drift probabilities from 0.25 to 1.0. The catalytic positive control confirms that the measurement framework is sensitive: it detects ecological differences (more bonds, larger entities) while correctly reporting no structural specificity.

Together, these results characterize a boundary condition for emergent complexity. Below this boundary, defined by low-expressiveness random local bonding rules with a 3-tuple context, entity assembly is fully explained by size. We emphasize that this conclusion is scoped to the rule class tested: uniform random bond probabilities conditioned on (self type, neighbor count, dominant type). Richer rule representations, structured rule sampling, or adaptation could produce different outcomes. The boundary extends beyond the base 3-tuple context: partner-specific rules (45-entry) and motif-gated catalysis also yield 0% excess (§, §), confirming that richer local conditioning alone does not suffice. Crossing this boundary likely requires biased rule sampling, environmental gradients, or temporal dynamics such as rule adaptation. Spatially-oriented rules remain an untested intermediate.

## Relation to Prior Work

The observation that random dynamics produce trivial structures resonates with classical results in cellular automata, where most random rules produce either homogeneous or chaotic behavior, with complex dynamics concentrated at the "edge of chaos" (Langton, 1990; Wolfram, 2002). Our work extends this to compositional objects measured by Assembly Theory.

Our null baseline connects to Lehman and Stanley's novelty search (Lehman and Stanley, 2011) and Taylor et al.'s open-ended evolution framework (Taylor et al., 2016; Taylor, 2015): our result characterizes the "floor" below which no selection mechanism is needed: random bonding alone produces a stable entity ecology without any pressure toward complexity. This establishes the minimal baseline against which objective-free interventions can be measured.

## Limitations

First, although our parameter sweep covers density, grid size, and drift strength, and we additionally test partner-specific rules (§), other rule representations (orientation-dependent rules, rule evolution, and structured sampling) remain untested. Second, the 500-step horizon, while sufficient for stationarity (Figure 9), may miss very slow transients in larger systems. Third, entity observations at each step are not fully independent (the same entity may persist across steps); our significance testing uses all observations but the 0% excess holds identically when restricted to unique types. Fourth, while we supplement the assembly index with automorphism counts and typed-motif census (§), a full multi-metric comparison incorporating compression-based complexity and information-theoretic measures is deferred to future work.

# Conclusion

Across 5,000 simulations and 7.1M entity observations, uniform random bonding rules produce a stable entity ecology but no structurally non-trivial assembly. The edge-removal identity $a_{\text{exact}}(G) = |E(G)|$ makes size-driving unsurprising under that formulation; the empirical finding is that the reuse-aware formulation, where $a_r(G) < |E(G)|$ when sub-object reuse applies, also shows zero excess. This result holds across parameter sweeps, partner-specific rules, and motif-gated catalysis. Within this rule class, structural bias appears necessary for emergent complexity even without fitness functions. Reporting this negative result is itself valuable (Fanelli, 2012), and the boundary condition provides a concrete baseline against which richer rule representations can be measured.


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